Software Requirements Specification (SRS)

Project: The Last Stronghold
Platform: Unreal Engine 5 (Blueprint-Only Implementation)
Prepared for: Software Engineering Capstone Review
Author: Christian Alvarez, Stanley Delva, Hendrick Christian
Document Type: IEEE-Style Software Requirements Specification
Version: 1.2
Date: 2026

Source-bounded documentation notice: This Software Requirements Specification is derived directly from the documented systems and actors described in the Software Design Specification for The Last Stronghold.

Revision History

Version	Date	Description
1.0	March 16 2026	Initial SRS creation
1.1	March 17 2026	Added diagrams and traceability
1.2	March 19 2026	Merged Capstone and Professor documentation

Table of Contents

1. Introduction
  1.1 Purpose
  1.2 Scope
  1.3 Definitions, Acronyms, and Abbreviations
  1.4 References

2. Overall Description
  2.1 Product Perspective
  2.2 Product Functions
  2.3 User Classes
  2.4 Operating Environment
  2.5 Design Constraints
  2.6 Assumptions and Dependencies

3. System Requirements
  3.1 Wave Management Requirements
  3.2 Player Character Requirements
  3.3 Enemy System Requirements
  3.4 Defense System Requirements
  3.5 Resource System Requirements
  3.6 Tower Objective Requirements
  3.7 User Interface Requirements

4. System Models
  4.1 System Architecture Diagram
  4.2 UML Class Relationship Overview

5. External Interface Requirements
6. Non-Functional Requirements
7. Requirements Traceability Matrix
8. Figures

1. Introduction

1.1 Purpose

This document defines the functional and non-functional requirements for The Last Stronghold. The SRS describes the system capabilities required to support gameplay, enemy wave progression, defensive structures, resource collection, and user interface feedback.

1.2 Scope

The Last Stronghold is a third-person action tower defense game built in Unreal Engine 5 using Blueprint visual scripting. Players defend a magical obelisk from waves of enemies using combat abilities and automated defenses.

1.3 Definitions

Term	Definition
Blueprint	Unreal Engine visual scripting system
GameMode	Controls gameplay rules and wave progression
HUD	Heads-up display presenting gameplay information
Soul Shards	Collectible currency used to build defenses
Turret	Automated defense structure that attacks enemies
Obelisk Tower	Primary objective that must be defended

1.4 References

IEEE Standard 1016-2009 – Software Design Descriptions
Unreal Engine 5 Documentation
The Last Stronghold Blueprint Breakdown Report

2. Overall Description

2.1 Product Perspective

The system is implemented as a modular Unreal Engine project composed of gameplay actors, UI widgets, and controller systems.

2.2 Product Functions

- Wave-based enemy spawning
- Player combat interaction
- Automated turret defense systems
- Resource collection and currency management
- Game win and loss determination
- User interface display updates

2.3 User Classes

User Type	Description
Player	Individual playing the game
Developer	Maintains and extends gameplay systems
Instructor	Evaluates the capstone project
Tester	Validates gameplay functionality

2.4 Operating Environment

Engine: Unreal Engine 5
Platform: PC
Implementation: Blueprint scripting
Input Devices: Keyboard and Mouse

2.5 Design Constraints

All gameplay systems must be implemented using Unreal Engine Blueprint visual scripting.

2.6 Assumptions and Dependencies

The Unreal Engine runtime environment must be available and assets must be imported correctly.

3. System Requirements

Requirement ID	Requirement Description
REQ-1	The system shall spawn enemy waves.
REQ-1.1	The system shall track active enemies.
REQ-1.2	The system shall initiate countdowns between waves.
REQ-2	The player character shall move through the level.
REQ-2.1	The player shall attack enemies.
REQ-3	Enemies shall move toward the tower objective.
REQ-3.1	Enemies shall enter a death state when health reaches zero.
REQ-4	Turrets shall detect enemies within range.
REQ-4.1	Turrets shall rotate toward enemies.
REQ-4.2	Turrets shall spawn attack actors.
REQ-5	Soul shard pickups shall grant currency.
REQ-5.1	Pickups shall destroy themselves after collection.
REQ-6	The tower shall receive damage from enemies.
REQ-6.1	The game shall end if the tower is destroyed.
REQ-7	The HUD shall display wave and currency information.
REQ-7.1	The game shall display win and loss screens.

4. System Models

4.1 System Architecture Diagram

```
+---------------------+
|    GameMode       |
| Wave Controller   |
+----------+----------+
        |
+----------+----------+
|                 |
v                 v
Player System     Enemy System
(Character)       (BP_Skeleton)
     |
     v
Defense Systems
(Turrets / Explosives)
     |
     v
Tower Objective
```

Figure 1 — High-level system architecture for The Last Stronghold.

4.2 UML Class Relationship Overview

```
[GameMode]
  |
  | manages
  v
[Enemy Actor] ---- attacked by ---- [Player Character]
    |
    | targeted by
    v
[Turret System]
    |
    v
[Explosion / Projectile]
```

Figure 2 — Simplified UML interaction model referencing REQ-3 and REQ-4.

7. Requirements Traceability Matrix

Requirement	Subsystem	Design Component
REQ-1	Wave System	GM_TowerDefense
REQ-2	Player System	PlayerCharacter
REQ-3	Enemy System	BP_Skeleton
REQ-4	Defense System	BP_Turret / BP_Turret_Blast
REQ-5	Resource System	BP_SoulShardPickup
REQ-6	Objective System	BP_Tower
REQ-7	UI System	W_MainHUD / W_BuyMenu / W_HealthBar

8. Figures

Figure 1 — High-level system architecture diagram.

Figure 2 — UML interaction overview.